

# Combining factors

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## Abstract

While the academic literature focuses on beta exposure, most practitioners apply characteristics-based scorings to obtain factor portfolios. This paper explores how firm-level characteristics can be combined for optimal factor portfolios. Portfolios encompassing multiple factors are less volatile and have higher after-cost returns than the market or single factor portfolios. We also demonstrate that buy- and sell-thresholds play a critical role in shaping portfolio return, risk, and turnover preferences. Our empirical findings reveal optimal weights for the combination of individual factors, though we acknowledge the 1/N portfolio as a challenging benchmark.

*JEL classification:* G11, G12, G15

*Keywords:* Portfolio construction, factor investing, transaction costs, investment strategies

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# 1 Introduction

The finance literature made substantial progress in identifying factors that drive stocks' risk and return characteristics. Building on Ross (1976) APT, numerous papers provide evidence that the cross-section of stock returns is driven by covariances with factors. Starting with Fama and French (1993), many of these factors have been defined as the return series of long-short portfolios, generated through double-sorting stocks by firm characteristics.

When it comes to implementing characteristics-based portfolio strategies, the academic literature offers only limited guidance. Arnott et al. (2005), weight firms based on "main-street" measures, such as revenue, book-equity, gross dividends, cash-flow and employment metrics. Asness et al. (2013) utilize rank-based value and momentum signals to construct factor portfolios. They show that the signal weighted factor portfolios, outperforms simple portfolio sort spreads. Fisher et al. (2016) investigate value and momentum characteristics score portfolios, selecting stocks based on high characteristic loadings rather than relying on high beta exposure.

However, none of the aforementioned literature investigates the combination of multiple characteristics to form long-only portfolios. Our analysis seeks to address this gap and provide valuable insights for the effective implementation of characteristics-based long-only portfolio strategies. First, we extend the factor set of Fisher et al. (2016) to the investment and profitability factor (see Fama and French, 2015). We find that equity portfolios based on asset-growth and gross profitability exhibit above market returns and Sharpe ratios, similar to portfolios formed on book-to-market. Second, portfolios constructed by balance sheet characteristics tend to change sluggish, resulting in low turnover and long holding-periods. Third, the combination of multiple characteristics ensures that the information conveyed by stock-level characteristics is multidimensional. These portfolios tend to outperform both the market and single-score portfolios. Fourth, we show that optimal weights for multivariate portfolios are almost equally distributed.

## 2 Univariate portfolios

Contrary to the academic paradigm of portfolio sorts, which neglects turnover and trading cost considerations, institutional investors may want to avoid divesting stocks that have marginally entered the nearby sell range. Fisher et al. (2016) addresses this by implementing a straightforward yet effective non-trading band between buy and sell thresholds. This involves calculating each stock's

characteristic score  $S_k$ , as the ratio of the market capitalization of all stocks with the same or lower characteristic values in the eligible universe divided by the total market capitalization of all stocks.<sup>1</sup> For example, a firm with a book-to-market ratio of 2 receives a score of 80, if 80 percent of the total stock market capitalization consisted of stocks with a book-to-market ratio less than or equal to 2.

Table 1 shows results for univariate factor portfolios. Small market depicts the value-weighted portfolio of all stocks with a lower market capitalization than the NYSE median. Large market comprises all stocks above the median. Transaction costs are defined as the observed bid-ask spread.<sup>2</sup> In the baseline portfolio, stocks with scores higher than 90 and not owned by the portfolio are purchased; firms with scores lower than 70, and owned by the portfolio, are sold. These portfolios produce after cost returns (R-TC) between 14.5% and 17.5% p.a. with Sharpe ratios (SR-TC) between 0.37 and 0.52. Portfolios built on balance sheet characteristics exhibit sluggish movement, evident through low turnover rates (Turn.) and long holding periods (T). Conversely, momentum requires significantly more turnover, leading to shorter holding periods.

Further, our analysis reveals that risk, return and trading properties are materially affected by the characteristics, but also by the chosen thresholds. Notably, for the value portfolio, the highest return is achieved using buy and sell thresholds of 89 and 57, respectively. A similar picture emerges when we look at optimal Sharpe ratios. The results for the momentum portfolio closely resemble the baseline thresholds for achieving optimal returns. However, when our objective is to maximize the portfolios' Sharpe ratio, optimal thresholds become remarkably lower and tighter. This results in a composition containing more securities and requiring less trading, effectively mitigating idiosyncratic risk and trading costs. Panel B unveils consistent results for large firms. Here, optimal thresholds for the momentum portfolio are significantly tighter; 95/87 for the maximum R-TC and 95/85 for the highest SR-TC portfolio, respectively. We attribute this observation to considerably lower transaction costs associated with the trading of large capitalization stocks.

### 3 Multivariate portfolios

Investors can easily gain exposure to multiple factors by allocating funds across sub-portfolios. This, however, disregards the covariance structure, and therefore risks associated with investing in the same firm through multiple channels. An alternative solution could be a best-in-class approach,

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<sup>1</sup>A detailed description of the characteristics is provided in the appendix.

<sup>2</sup>For details, see appendix.

wherein firms must be present in each top portfolio to be considered. In light of the exceedingly low probabilities of firms concurrently belonging to each top portfolio, the feasibility of this strategy is contingent upon a substantial number of investable firms. To address both concerns, a viable approach involves the combination of multiple characteristics to an aggregate score. This way, the portfolio comprises of a sufficiently large number of stocks which in turn are influenced by multiple favorable characteristics.

DeMiguel et al. (2009) propose a  $1/N$  naive-diversification as an obvious benchmark. We therefore start with equally weighting all scores, denoted as “Avg. V/I/P/M” in Table 2. The resulting portfolio exhibits impressive results, outperforming the market and all univariate portfolios with an annualized return and Sharpe ratio of 18.4% and 0.63, respectively. While the incorporation of momentum leads to an increase in turnover, associated transaction costs are more than offset by higher returns. A portfolio that buys stocks with scores above 92 and sells those below 78, exhibits the highest return (19.1%), with a Sharpe ratio of 0.65.

Albeit smaller, we find consistent results for large firms. The baseline portfolio produces an excess-market return of 1.8%, with similar volatility, translating to a Sharpe ratio of 0.54. Maximizing the return (15.3% p.a.) necessitates extremely narrow thresholds, restricting the portfolio to just 52 securities and requiring almost 290% turnover. To maximize the Sharpe ratio, the sell threshold is lowered to 74, expanding the no-trade range, increasing the number of stocks, and reducing trading activity by half.

### 3.1 Conditional scoring

Several solutions potentially reduce portfolio in- and outflows. For example, investors could use wider no-trading ranges. Alternatively, firms already owned could be augmented by retention-terms. Further, sophisticated models could evaluate the expected gain compared to a certain trading cost.

In this paper, we use momentum as a “conditioning” threshold to the existing strategy. Utilizing momentum serves two purposes. First, adjustments influenced by momentum exhibit a notably swifter integration compared to those reliant on balance sheet metrics. Second, the strategy systematically avoids the acquisition of underperforming stocks and the divestment of outperforming stocks, regardless of the dependent characteristic. We therefore exclude securities with a momentum score below 50 from purchase considerations, and exempt those with a score exceeding 50 from selling. Setting the threshold-score at 50 ensures a sufficiently large and diversified portfolio.

“Avg. V/I/P | Mom  $\geq 50$ ” in Table 2 reports the results. Compared to the unconditional portfolio, the small baseline model produces a 20 basis point lower annualized return of 18.2%, equivalent Sharpe ratio of 0.63, but requires only half the turnover. The next two lines present optimal return and Sharpe ratio portfolios; both produce returns of 18.4%, Sharpe ratios of 0.65 and require merely 50% turnover per year.

Utilizing a conditional momentum threshold for large firms results in elevated returns, increased Sharpe ratios and reduced turnover.

## 4 Optimal weights

In this section, we determine optimal combination of characteristic weights that maximize the portfolios after cost Sharpe ratio. We define scores for stock  $i$  as:

$$\begin{aligned} SCORE_{i,t} = & w_{t,1} * SCORE_{i,t}^{Value} + \\ & w_{t,2} * SCORE_{i,t}^{Investment} + \\ & w_{t,3} * SCORE_{i,t}^{Profitability}, \end{aligned} \tag{1}$$

and buy (sell) only if the combined score is above (below) 90 (70) conditional that  $SCORE_{i,t}^{Momentum}$  is also above (below) 50.

Figure 1 shows heat-maps for small and large stocks. Moving along the axes, the assigned weight increases in value, investment and profitability, with corner allocations being identical to the univariate portfolios. We again find overwhelming evidence that diversified portfolios exhibit higher Sharpe ratios than concentrated portfolios. The distribution of weights across multiple factors allows for mitigation of potential impact of extreme movements in any individual characteristic. The sign represents optimal weights: 36%, 28%, and 36% for small, and 19%, 50%, and 31% for large value, investment, and profitability characteristic scores, respectively. For the out-of-sample results, we estimate the parameters using an expanding window, where only data up to time  $t$  are considered and a burn-in period of 120 months is applied to obtain initial estimates.<sup>3</sup> Detailed performance statistics are reported in Table 2.

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<sup>3</sup>We provide out-of-sample weights in the appendix.

## 5 Conclusion

This paper challenges the prevailing focus on beta-exposure as a measure of expected returns and instead supports characteristic-based scoring to construct long-only factor portfolios. We find strong empirical evidence that the nature of the characteristic plays a crucial role in shaping portfolio dynamics, and therefore, estimate optimal thresholds to account for this variability. By carefully selecting appropriate thresholds that align with the unique properties of the underlying characteristics, investors can effectively manage portfolio turnover and transaction costs, ultimately enhancing overall performance metrics. In addition, our paper suggests that the combination of multiple characteristics dominates univariate portfolios. Thus, we advocate for the incorporation of slow-moving attributes in conjunction with momentum, as this combination has the potential for excess returns while mitigating portfolio turnover and transaction costs. Our analysis also derives optimal weights for multivariate factor portfolios. We find that allocating nearly equal weights to all characteristics results in the highest Sharpe ratios. Finally, our approach survives rigorous robustness checks encompassing sub-periods, various weighting schemes, and an analysis for Europe.

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Table 1: Univariate factor portfolios

|                                      |            | Buy | Sell | R-TC | SD   | SR-TC | Turn. | T  | N    |
|--------------------------------------|------------|-----|------|------|------|-------|-------|----|------|
| Panel A: Small capitalization stocks |            |     |      |      |      |       |       |    |      |
| Small market                         |            |     |      | 13.0 | 21.3 | 0.37  | 32.2  | 80 | 3892 |
| Value                                | Baseline   | 90  | 70   | 16.5 | 21.0 | 0.52  | 48.3  | 41 | 767  |
|                                      | Max. R-TC  | 89  | 57   | 16.6 | 20.5 | 0.53  | 42.1  | 48 | 894  |
|                                      | Max. SR-TC | 87  | 58   | 16.5 | 20.3 | 0.54  | 42.7  | 47 | 965  |
| Investment                           | Baseline   | 90  | 70   | 14.5 | 24.8 | 0.37  | 79.4  | 21 | 676  |
|                                      | Max. R-TC  | 61  | 50   | 15.8 | 20.5 | 0.50  | 67.4  | 25 | 1478 |
|                                      | Max. SR-TC | 56  | 50   | 15.8 | 20.4 | 0.50  | 68.3  | 25 | 1549 |
| Profitability                        | Baseline   | 90  | 70   | 15.1 | 22.8 | 0.43  | 58.7  | 30 | 333  |
|                                      | Max. R-TC  | 90  | 82   | 15.5 | 23.0 | 0.43  | 67.5  | 24 | 287  |
|                                      | Max. SR-TC | 52  | 50   | 14.6 | 20.7 | 0.45  | 56.0  | 30 | 1121 |
| Momentum                             | Baseline   | 90  | 70   | 17.5 | 25.4 | 0.46  | 173.1 | 7  | 467  |
|                                      | Max. R-TC  | 90  | 71   | 17.7 | 25.4 | 0.47  | 176.5 | 7  | 461  |
|                                      | Max. SR-TC | 65  | 51   | 16.6 | 21.2 | 0.52  | 159.0 | 7  | 1178 |
| Panel B: Large capitalization stocks |            |     |      |      |      |       |       |    |      |
| Large market                         |            |     |      | 11.7 | 15.5 | 0.43  | 8.4   | 59 | 925  |
| Value                                | Baseline   | 90  | 70   | 12.9 | 15.3 | 0.50  | 24.5  | 40 | 195  |
|                                      | Max. R-TC  | 95  | 80   | 13.4 | 16.5 | 0.49  | 31.0  | 35 | 117  |
|                                      | Max. SR-TC | 91  | 70   | 13.1 | 15.3 | 0.51  | 24.3  | 41 | 186  |
| Investment                           | Baseline   | 90  | 70   | 13.6 | 15.7 | 0.53  | 59.7  | 16 | 154  |
|                                      | Max. R-TC  | 92  | 75   | 14.2 | 16.1 | 0.55  | 64.5  | 15 | 125  |
|                                      | Max. SR-TC | 80  | 70   | 13.5 | 15.0 | 0.55  | 61.2  | 14 | 223  |
| Profitability                        | Baseline   | 90  | 70   | 11.3 | 15.9 | 0.39  | 24.2  | 36 | 97   |
|                                      | Max. R-TC  | 72  | 56   | 12.8 | 15.5 | 0.49  | 20.4  | 37 | 219  |
|                                      | Max. SR-TC | 68  | 56   | 12.7 | 15.4 | 0.49  | 21.2  | 35 | 239  |
| Momentum                             | Baseline   | 90  | 70   | 15.7 | 20.0 | 0.50  | 170.5 | 7  | 181  |
|                                      | Max. R-TC  | 95  | 87   | 18.4 | 22.9 | 0.55  | 236.6 | 5  | 98   |
|                                      | Max. SR-TC | 95  | 85   | 18.3 | 22.8 | 0.55  | 219.2 | 5  | 102  |

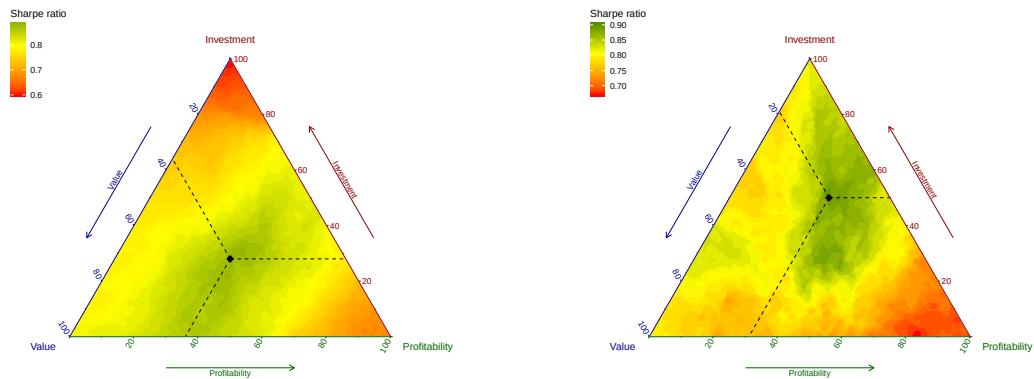
This table shows results for univariate value, investment, profitability and momentum factor portfolios. Baseline reports portfolios with buy thresholds at 90 and sell thresholds at 70. Max. R-TC and Max. SR-TC indicate the optimal buy and sell thresholds for portfolios with maximum return and Sharpe ratio after transaction costs, respectively. Columns depict the annualized after transaction cost return, standard deviation, Sharpe ratio, turnover, average monthly holding period, and average number of stocks in the portfolio.

Table 2: Multivariate factor portfolios

|                                      |               | Buy | Sell | R-TC | SD   | SR-TC | Turn. | T  | N    |
|--------------------------------------|---------------|-----|------|------|------|-------|-------|----|------|
| Panel A: Small capitalization stocks |               |     |      |      |      |       |       |    |      |
| Small market                         |               |     |      | 13.0 | 21.3 | 0.37  | 32.2  | 80 | 3892 |
| Avg. V/I/P/M                         | Baseline      | 90  | 70   | 18.4 | 19.8 | 0.63  | 113.2 | 11 | 443  |
|                                      | Max. R-TC     | 92  | 78   | 19.1 | 20.1 | 0.65  | 132.4 | 10 | 343  |
|                                      | Max. SR-TC    | 92  | 72   | 19.0 | 20.0 | 0.65  | 113.9 | 11 | 378  |
| Avg. V/I/P   Mom $\geq 50$           | Baseline      | 90  | 70   | 18.2 | 19.5 | 0.63  | 56.9  | 28 | 403  |
|                                      | Max. R-TC     | 94  | 56   | 18.4 | 19.4 | 0.65  | 45.4  | 40 | 343  |
|                                      | Max. SR-TC    | 91  | 58   | 18.4 | 19.3 | 0.65  | 46.9  | 36 | 441  |
| Optimal V/I/P   Mom $\geq 50$        | In-sample     | 90  | 70   | 19.3 | 18.8 | 0.72  | 51.8  | 30 | 437  |
|                                      | Out-of-sample | 90  | 70   | 19.1 | 18.9 | 0.71  | 49.8  | 31 | 447  |
| Panel B: Large capitalization stocks |               |     |      |      |      |       |       |    |      |
| Large market                         |               |     |      | 11.7 | 15.5 | 0.43  | 8.4   | 59 | 925  |
| Avg. V/I/P/M                         | Baseline      | 90  | 70   | 13.5 | 15.4 | 0.54  | 117.9 | 9  | 137  |
|                                      | Max. R-TC     | 95  | 94   | 15.3 | 17.4 | 0.56  | 289.0 | 4  | 52   |
|                                      | Max. SR-TC    | 94  | 74   | 14.8 | 16.2 | 0.58  | 122.7 | 9  | 97   |
| Avg. V/I/P   Mom $\geq 50$           | Baseline      | 90  | 70   | 14.1 | 15.4 | 0.57  | 48.5  | 23 | 119  |
|                                      | Max. R-TC     | 95  | 95   | 15.3 | 17.7 | 0.55  | 96.6  | 15 | 46   |
|                                      | Max. SR-TC    | 94  | 74   | 15.2 | 16.0 | 0.61  | 51.8  | 23 | 77   |
| Optimal V/I/P   Mom $\geq 50$        | In-sample     | 90  | 70   | 15.7 | 14.8 | 0.70  | 55.4  | 21 | 109  |
|                                      | Out-of-sample | 90  | 70   | 14.6 | 14.8 | 0.63  | 45.0  | 24 | 154  |

This table shows performance statistics for multivariate portfolios, conditional portfolios and optimal weighted portfolios. Baseline reports portfolios with buy thresholds at 90 and sell thresholds at 70. Max. R-TC and Max. SR-TC indicate the optimal buy and sell thresholds for portfolios with maximum return and Sharpe ratio after transaction costs, respectively. Columns depict the annualized after transaction cost return, standard deviation, Sharpe ratio, turnover, average monthly holding period, and average number of stocks in the portfolio.

Figure 1: Optimal in-sample weights for conditional multivariate portfolios  
 (a) Small capitalization stocks  
 (b) Large capitalization stocks



This figure shows heatmaps for the optimal combinations of factor scores for the conditional scoring approach. The black dot shows the optimal score combination.

# Appendices

## A Data

Our empirical analysis builds on data from CRSP, and covers all ordinary common shares (share-code 10 and 11) traded on NYSE, AMEX, and Nasdaq during the period from January 1965 to June 2020. Accounting data are obtained from Compustat and lagged by six months (see Fama and French, 1993). The risk-free rate (i.e. the one-month Treasury bill rate) is obtained from Kenneth French's website

We consider the characteristics (i) size (market capitalization) (ii) value (book-to-market ratio) (iii) investment (change in total assets) (iv) profitability (revenue minus cost of goods sold, interest expense, and selling, general, and administrative expense divided by the sum of book equity and minority interest) and (v) momentum (return from month  $t-12$  to month  $t-2$ ). When assigning these characteristics to a stock, we follow Fama and French (1993, 2015) and keep balance sheet items constant for one year. Further, we divide the universe of stocks at the median market capitalization of all NYSE stocks.

## B Transaction costs

Transaction costs are calculated as  $(\text{price ask} - \text{price bid})/\text{price ask}$ . If bid or ask quotes are missing, we use the previous months bid-ask spread. In case this information is missing as well, we assign transaction costs from a group of firms in the same size decile. Figure B.1 displays round-trip transaction costs for small and large firms over time.

Figure B.1: Transaction costs

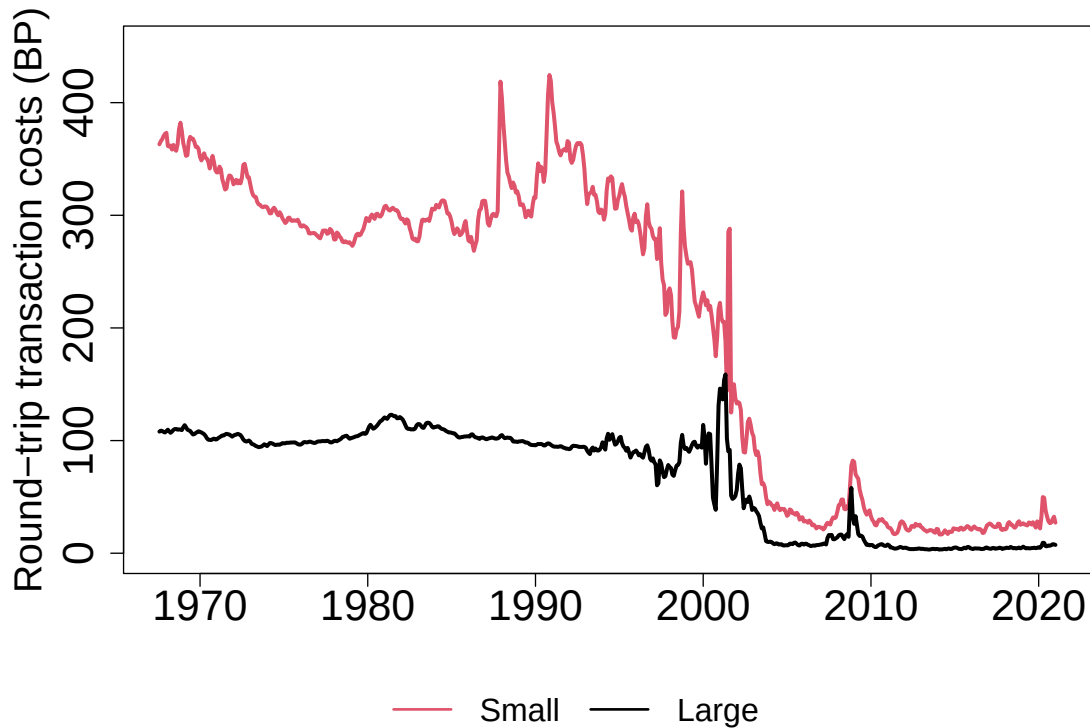


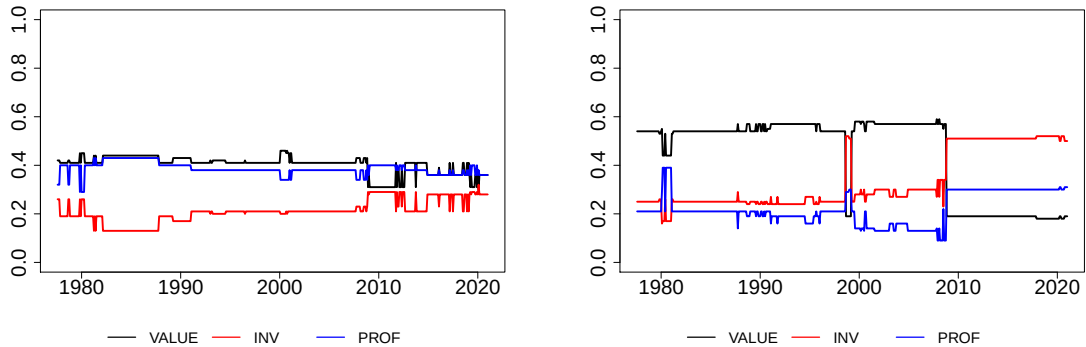
Figure B.1 shows the round-trip transaction costs for small and large stocks. Small stocks are those below the NYSE median market capitalization. Large stocks are those above the NYSE market capitalization. Transaction costs are calculated as  $(\text{price ask} - \text{price bid})/\text{price ask}$ . If bid or ask quotes are missing, we use the most recent bid-ask spread. In case this information is missing as well, we assign transaction costs from a group of firms in the same size decile. The sample period is July 1967 to June 2020.

## C Out-of-sample weights

Figure C.1: Optimal out-of-sample weights for conditional multivariate portfolios

(a) Small capitalization stocks

(b) Large capitalization stocks



These figures display optimal out-of-sample characteristic weights for multivariate factor portfolios. The sample period is July 1977 to June 2020.

## D Robustness analysis

We perform a wide range of robustness checks. First, in Table D.1 we investigate sub-periods of our 5 decade long data set. That way, we are able to control for different economic regimes, structural changes in transaction costs as well as post-publication bias that results from choosing a specific factor set. Second, in Table D.2 we deviate from our framework and invest equally into the top 100 stocks with the highest score, given they satisfy our buy and sell threshold boundaries. Third, in Tables D.3, D.4 and D.5 we test our model completely out-of-sample in Europe.<sup>4</sup>

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<sup>4</sup>The dataset consists of 16 European countries: Austria, Belgium, Denmark, Finland, France, Germany, Greece, Ireland, Italy, the Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, and the United Kingdom. The sample of monthly observations covers 13,871 individual firms, from January 1980 to June 2018. The return data is obtained from Datastream, whereas the accounting data is from Worldscope. All returns are in U.S. dollars, include dividends and capital gains, and are not continuously compounded. The risk-free rate is the U.S. one month T-bill rate. The small sample consists of the lowest 10 percent of market capitalization, representing roughly 3550 firms. Large firms make up 90 percent of market capitalization, with 1270 firms on average in the sample. Due to bad data quality for the profitability factor (specifically for cost of goods sold, administrative expenses and interest expenses) we deviate from United States definition and replace gross profit by net income. Therefore, our profitability factor is defined as net income of the end of the previous fiscal year, divided by the book equity of the end of previous fiscal year.

Table D.1: Factor portfolios in different sub-periods

|                                      |                            | R-TC | SD   | SR-TC | Turn. | T  | N    |
|--------------------------------------|----------------------------|------|------|-------|-------|----|------|
| Panel A: Small capitalization stocks |                            |      |      |       |       |    |      |
| 1970-1979                            | Small Market               | 12.9 | 24.1 | 0.24  | 67.9  | 52 | 2204 |
|                                      | Value                      | 18.6 | 25.1 | 0.43  | 89.8  | 33 | 502  |
|                                      | Investment                 | 16.2 | 27.0 | 0.33  | 150.1 | 19 | 423  |
|                                      | Profitability              | 13.8 | 27.5 | 0.24  | 119.7 | 25 | 270  |
|                                      | Momentum                   | 15.1 | 24.1 | 0.33  | 338.5 | 7  | 395  |
|                                      | Avg. V/I/P/M               | 18.9 | 22.8 | 0.49  | 216.6 | 11 | 331  |
| 1980-1989                            | Avg. V/I/P   Mom $\geq 50$ | 19.0 | 22.4 | 0.50  | 114.2 | 23 | 301  |
|                                      | Small Market               | 16.0 | 19.3 | 0.32  | 67.5  | 48 | 3367 |
|                                      | Value                      | 21.6 | 16.7 | 0.66  | 95.2  | 31 | 694  |
|                                      | Investment                 | 13.5 | 20.7 | 0.20  | 146.7 | 19 | 772  |
|                                      | Profitability              | 17.6 | 21.7 | 0.35  | 121.4 | 22 | 347  |
|                                      | Momentum                   | 20.1 | 23.4 | 0.42  | 335.0 | 7  | 543  |
| 1990-1999                            | Avg. V/I/P/M               | 26.0 | 18.5 | 0.79  | 213.0 | 11 | 421  |
|                                      | Avg. V/I/P   Mom $\geq 50$ | 26.2 | 17.4 | 0.85  | 115.9 | 23 | 357  |
|                                      | Small Market               | 15.1 | 17.0 | 0.54  | 80.7  | 47 | 4373 |
|                                      | Value                      | 16.9 | 15.3 | 0.70  | 101.3 | 31 | 1061 |
|                                      | Investment                 | 17.6 | 19.8 | 0.57  | 158.7 | 19 | 934  |
|                                      | Profitability              | 18.0 | 17.3 | 0.68  | 120.8 | 24 | 445  |
| 2000-2009                            | Momentum                   | 28.5 | 24.2 | 0.84  | 329.9 | 7  | 610  |
|                                      | Avg. V/I/P/M               | 23.3 | 14.8 | 1.09  | 213.4 | 11 | 573  |
|                                      | Avg. V/I/P   Mom $\geq 50$ | 20.7 | 13.8 | 1.01  | 121.2 | 23 | 476  |
|                                      | Small Market               | 9.0  | 23.4 | 0.25  | 80.6  | 50 | 3699 |
|                                      | Value                      | 12.8 | 23.0 | 0.40  | 105.9 | 30 | 931  |
|                                      | Investment                 | 9.0  | 32.3 | 0.18  | 175.4 | 19 | 746  |
| 2010-2020                            | Profitability              | 12.1 | 22.1 | 0.39  | 118.6 | 26 | 351  |
|                                      | Momentum                   | 5.0  | 31.2 | 0.06  | 367.3 | 7  | 484  |
|                                      | Avg. V/I/P/M               | 12.3 | 19.4 | 0.46  | 236.7 | 11 | 550  |
|                                      | Avg. V/I/P   Mom $\geq 50$ | 12.4 | 18.8 | 0.48  | 119.9 | 25 | 491  |
|                                      | Small Market               | 15.9 | 20.8 | 0.68  | 68.3  | 58 | 2523 |
|                                      | Value                      | 13.9 | 23.2 | 0.54  | 97.3  | 37 | 661  |
|                                      | Investment                 | 15.9 | 24.1 | 0.59  | 165.0 | 20 | 548  |
|                                      | Profitability              | 17.4 | 23.6 | 0.66  | 111.6 | 29 | 239  |
|                                      | Momentum                   | 19.6 | 22.8 | 0.76  | 346.0 | 7  | 340  |
|                                      | Avg. V/I/P/M               | 15.1 | 21.6 | 0.63  | 236.5 | 11 | 376  |
|                                      | Avg. V/I/P   Mom $\geq 50$ | 15.2 | 22.8 | 0.60  | 113.7 | 27 | 332  |
| Panel B: Large capitalization stocks |                            |      |      |       |       |    |      |
| 1970-1979                            | Large Market               | 7.7  | 16.2 | 0.07  | 13.8  | 54 | 809  |
|                                      | Value                      | 11.8 | 16.2 | 0.31  | 50.7  | 36 | 171  |
|                                      | Investment                 | 12.8 | 17.6 | 0.34  | 119.3 | 18 | 150  |
|                                      | Profitability              | 5.4  | 17.1 | -0.05 | 35.3  | 33 | 69   |
|                                      | Momentum                   | 11.5 | 17.6 | 0.27  | 324.1 | 8  | 168  |
|                                      | Avg. V/I/P/M               | 13.2 | 15.6 | 0.40  | 207.9 | 11 | 127  |
| 1980-1989                            | Avg. V/I/P   Mom $\geq 50$ | 14.3 | 16.7 | 0.43  | 87.7  | 26 | 114  |
|                                      | Large Market               | 17.7 | 16.4 | 0.47  | 22.3  | 47 | 834  |
|                                      | Value                      | 19.7 | 14.5 | 0.66  | 60.8  | 34 | 132  |
|                                      | Investment                 | 21.1 | 16.5 | 0.64  | 120.4 | 17 | 133  |
|                                      | Profitability              | 17.0 | 18.3 | 0.39  | 64.4  | 26 | 83   |
|                                      | Momentum                   | 19.6 | 21.1 | 0.44  | 357.6 | 7  | 151  |
| 1990-1999                            | Avg. V/I/P/M               | 20.5 | 18.3 | 0.56  | 222.9 | 10 | 103  |
|                                      | Avg. V/I/P   Mom $\geq 50$ | 21.2 | 17.4 | 0.62  | 90.3  | 23 | 94   |
|                                      | Large Market               | 20.3 | 13.4 | 1.03  | 21.7  | 37 | 1069 |
|                                      | Value                      | 17.0 | 13.0 | 0.83  | 56.0  | 30 | 225  |
|                                      | Investment                 | 17.3 | 12.1 | 0.92  | 115.4 | 16 | 158  |
|                                      | Profitability              | 18.7 | 14.7 | 0.84  | 51.6  | 29 | 90   |
| 2000-2009                            | Momentum                   | 32.2 | 20.0 | 1.16  | 310.1 | 7  | 179  |
|                                      | Avg. V/I/P/M               | 20.0 | 13.7 | 0.98  | 219.7 | 9  | 132  |
|                                      | Avg. V/I/P   Mom $\geq 50$ | 17.7 | 12.7 | 0.90  | 93.2  | 22 | 110  |
|                                      | Large Market               | 0.8  | 16.2 | -0.12 | 21.0  | 35 | 1000 |
|                                      | Value                      | 6.8  | 16.4 | 0.23  | 55.6  | 29 | 258  |
|                                      | Investment                 | 5.5  | 16.6 | 0.16  | 133.3 | 14 | 185  |
| 2010-2020                            | Profitability              | 1.7  | 14.4 | -0.07 | 60.0  | 28 | 112  |
|                                      | Momentum                   | 0.1  | 23.9 | -0.11 | 348.0 | 7  | 225  |
|                                      | Avg. V/I/P/M               | 2.8  | 13.6 | -0.00 | 255.6 | 9  | 181  |
|                                      | Avg. V/I/P   Mom $\geq 50$ | 5.3  | 14.8 | 0.16  | 126.7 | 21 | 144  |
|                                      | Large Market               | 16.0 | 14.3 | 1.01  | 13.9  | 51 | 946  |
|                                      | Value                      | 13.0 | 17.2 | 0.68  | 35.9  | 37 | 163  |
|                                      | Investment                 | 15.5 | 15.1 | 0.92  | 112.9 | 16 | 143  |
|                                      | Profitability              | 15.4 | 14.2 | 0.97  | 41.0  | 38 | 107  |
|                                      | Momentum                   | 20.0 | 17.5 | 1.02  | 342.7 | 7  | 177  |
|                                      | Avg. V/I/P/M               | 15.9 | 15.0 | 0.95  | 257.8 | 8  | 143  |
|                                      | Avg. V/I/P   Mom $\geq 50$ | 18.8 | 15.5 | 1.08  | 108.5 | 21 | 109  |

This table shows performance statistics for factor portfolios during different sub periods. Columns depict the annualized after transaction cost return, standard deviation, Sharpe ratio, turnover, average monthly holding period, and average number of stocks in the portfolio.

Table D.2: Factor portfolios with limited portfolio size

|                                      | N = unlimited |      |       |       |     | N = 100 |      |       |       |    |
|--------------------------------------|---------------|------|-------|-------|-----|---------|------|-------|-------|----|
|                                      | R-TC          | SD   | SR-TC | Turn. | T   | R-TC    | SD   | SR-TC | Turn. | T  |
| Panel A: Small capitalization stocks |               |      |       |       |     |         |      |       |       |    |
| Small market                         | 13.7          | 22.5 | 0.37  | 132.8 | 770 | 15.0    | 23.5 | 0.40  | 127.4 | 66 |
| Value                                | 17.8          | 21.6 | 0.55  | 156.2 | 41  | 17.7    | 24.3 | 0.49  | 154.6 | 41 |
| Investment                           | 16.8          | 27.1 | 0.41  | 228.8 | 21  | 14.4    | 32.6 | 0.27  | 259.3 | 18 |
| Profitability                        | 14.1          | 23.4 | 0.37  | 173.0 | 30  | 14.0    | 24.6 | 0.35  | 164.0 | 32 |
| Momentum                             | 15.7          | 24.6 | 0.41  | 453.6 | 7   | 11.2    | 29.0 | 0.21  | 410.2 | 8  |
| Avg. V/M/I/P                         | 19.7          | 20.0 | 0.68  | 297.3 | 11  | 21.1    | 20.9 | 0.71  | 230.6 | 16 |
| Avg. V/I/P   Mom $\geq$ 50           | 19.7          | 19.5 | 0.70  | 148.2 | 34  | 20.3    | 20.9 | 0.67  | 130.6 | 39 |
| Panel B: Large capitalization stocks |               |      |       |       |     |         |      |       |       |    |
| Large market                         | 12.3          | 17.5 | 0.40  | 100.4 | 60  | 13.0    | 18.3 | 0.42  | 132.9 | 22 |
| Value                                | 14.0          | 16.7 | 0.52  | 114.4 | 40  | 14.6    | 16.7 | 0.54  | 93.2  | 56 |
| Investment                           | 14.0          | 17.5 | 0.49  | 191.2 | 16  | 14.2    | 18.0 | 0.49  | 181.5 | 17 |
| Profitability                        | 13.1          | 18.3 | 0.42  | 120.6 | 36  | 13.0    | 18.3 | 0.42  | 117.2 | 36 |
| Momentum                             | 16.6          | 21.7 | 0.50  | 414.0 | 7   | 17.2    | 23.9 | 0.47  | 349.7 | 7  |
| Avg. V/M/I/P                         | 14.7          | 16.5 | 0.56  | 302.9 | 9   | 15.3    | 16.8 | 0.58  | 263.0 | 10 |
| Avg. V/I/P   Mom $\geq$ 50           | 15.5          | 18.5 | 0.53  | 130.1 | 39  | 15.8    | 18.8 | 0.54  | 120.1 | 39 |

This table shows performance statistics for factor portfolios with a limited portfolio size. Columns depict the annualized after transaction cost return, standard deviation, Sharpe ratio, turnover, average monthly holding period, and average number of stocks in the portfolio.

## D.1 Europe

Figure D.1: Transaction costs

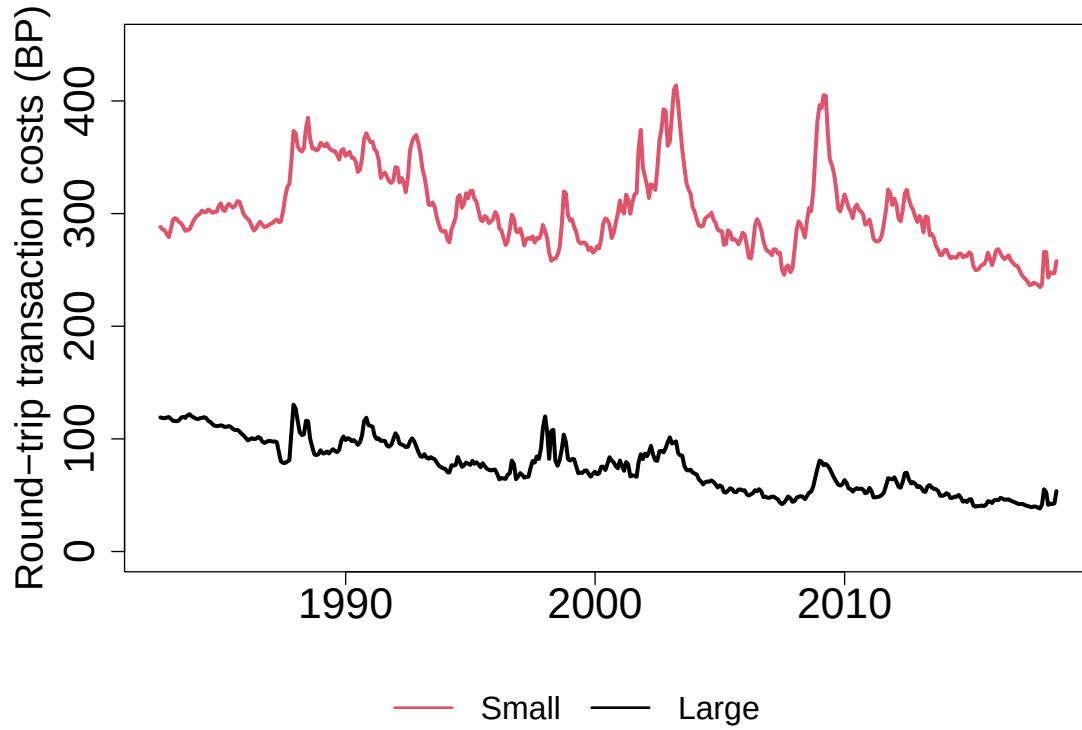


Figure D.1 shows the round-trip transaction costs for small and large stocks in Europe. Small stocks consists of the lowest 10 percent of market capitalization. Large stocks make up 90 percent of market capitalization. Transaction costs are calculated as  $(\text{price ask} - \text{price bid}) / \text{price ask}$ . If bid or ask quotes are missing, we use the most recent bid-ask spread. In case this information is missing as well, we assign transaction costs from a group of firms in the same size decile. The sample period is January 1980 to June 2018.

Table D.3: Univariate, multivariate and conditional factor portfolios in Europe

|                                      |            | Buy | Sell | R-TC | SD   | SR-TC | Turn. | T   | N    |
|--------------------------------------|------------|-----|------|------|------|-------|-------|-----|------|
| Panel A: Small capitalization stocks |            |     |      |      |      |       |       |     |      |
| Small market                         |            |     |      | 13.2 | 17.0 | 0.51  | 34.4  | 106 | 3542 |
| Value                                | Baseline   | 90  | 70   | 17.6 | 17.6 | 0.71  | 46.7  | 50  | 802  |
|                                      | Max. R-TC  | 95  | 50   | 18.0 | 17.7 | 0.73  | 37.0  | 68  | 663  |
|                                      | Max. SR-TC | 85  | 50   | 17.7 | 17.0 | 0.74  | 37.4  | 69  | 1199 |
| Investment                           | Baseline   | 90  | 70   | 13.1 | 17.8 | 0.48  | 82.4  | 23  | 738  |
|                                      | Max. R-TC  | 72  | 50   | 15.1 | 17.0 | 0.61  | 69.0  | 29  | 1463 |
|                                      | Max. SR-TC | 61  | 50   | 15.0 | 16.9 | 0.61  | 71.7  | 28  | 1677 |
| Profitability                        | Baseline   | 90  | 70   | 14.8 | 18.0 | 0.56  | 56.5  | 32  | 432  |
|                                      | Max. R-TC  | 60  | 59   | 15.2 | 17.5 | 0.60  | 59.7  | 31  | 1089 |
|                                      | Max. SR-TC | 60  | 59   | 15.2 | 17.5 | 0.60  | 59.7  | 31  | 1089 |
| Momentum                             | Baseline   | 90  | 70   | 23.1 | 19.6 | 0.88  | 157.2 | 7   | 553  |
|                                      | Max. R-TC  | 95  | 87   | 24.5 | 23.0 | 0.80  | 208.4 | 6   | 280  |
|                                      | Max. SR-TC | 90  | 69   | 23.0 | 19.6 | 0.88  | 154.5 | 8   | 560  |
| Avg. V/I/P/M                         | Baseline   | 90  | 70   | 23.3 | 17.7 | 0.98  | 111.5 | 11  | 487  |
|                                      | Max. R-TC  | 95  | 91   | 25.6 | 18.5 | 1.04  | 194.3 | 7   | 208  |
|                                      | Max. SR-TC | 95  | 91   | 25.6 | 18.5 | 1.04  | 194.3 | 7   | 208  |
| Avg. V/I/P   Mom $\geq$ 50           | Baseline   | 90  | 70   | 20.8 | 17.4 | 0.88  | 62.4  | 28  | 480  |
|                                      | Max. R-TC  | 95  | 95   | 22.8 | 18.2 | 0.93  | 91.5  | 17  | 186  |
|                                      | Max. SR-TC | 95  | 95   | 22.8 | 18.2 | 0.93  | 91.5  | 17  | 186  |
| Panel B: Large capitalization stocks |            |     |      |      |      |       |       |     |      |
| Large market                         |            |     |      | 11.2 | 15.3 | 0.45  | 6.2   | 91  | 1265 |
| Value                                | Baseline   | 90  | 70   | 14.7 | 19.4 | 0.52  | 26.3  | 42  | 207  |
|                                      | Max. R-TC  | 95  | 73   | 15.5 | 19.8 | 0.54  | 27.9  | 42  | 133  |
|                                      | Max. SR-TC | 92  | 73   | 15.5 | 19.5 | 0.55  | 27.8  | 40  | 178  |
| Investment                           | Baseline   | 90  | 70   | 14.4 | 17.7 | 0.55  | 67.1  | 15  | 144  |
|                                      | Max. R-TC  | 94  | 74   | 14.9 | 18.7 | 0.55  | 70.3  | 15  | 98   |
|                                      | Max. SR-TC | 73  | 69   | 14.3 | 17.0 | 0.57  | 68.6  | 14  | 267  |
| Profitability                        | Baseline   | 90  | 70   | 13.5 | 16.2 | 0.55  | 23.9  | 38  | 107  |
|                                      | Max. R-TC  | 85  | 83   | 14.6 | 16.5 | 0.61  | 33.9  | 25  | 112  |
|                                      | Max. SR-TC | 85  | 83   | 14.6 | 16.5 | 0.61  | 33.9  | 25  | 112  |
| Momentum                             | Baseline   | 90  | 70   | 15.7 | 19.3 | 0.57  | 178.5 | 7   | 183  |
|                                      | Max. R-TC  | 95  | 94   | 19.1 | 21.4 | 0.65  | 371.3 | 4   | 77   |
|                                      | Max. SR-TC | 95  | 91   | 18.8 | 21.0 | 0.65  | 297.9 | 5   | 88   |
| Avg. V/I/P/M                         | Baseline   | 90  | 70   | 17.1 | 17.6 | 0.69  | 121.3 | 10  | 138  |
|                                      | Max. R-TC  | 95  | 93   | 19.1 | 19.5 | 0.71  | 261.4 | 5   | 56   |
|                                      | Max. SR-TC | 95  | 82   | 18.8 | 18.2 | 0.75  | 151.8 | 8   | 76   |
| Avg. V/I/P   Mom $\geq$ 50           | Baseline   | 90  | 70   | 15.8 | 17.8 | 0.62  | 57.1  | 22  | 118  |
|                                      | Max. R-TC  | 93  | 89   | 17.4 | 18.8 | 0.66  | 77.2  | 17  | 72   |
|                                      | Max. SR-TC | 76  | 74   | 16.2 | 17.1 | 0.67  | 64.1  | 18  | 208  |

This table shows results for univariate value, investment, profitability and momentum factor portfolios in Europe. Baseline reports portfolios with buy thresholds at 90 and sell thresholds at 70. Max. R-TC and Max. SR-TC indicate the optimal buy and sell thresholds for portfolios with maximum return and Sharpe ratio after transaction costs, respectively. Columns depict the annualized after transaction cost return, standard deviation, Sharpe ratio, turnover, average monthly holding period, and average number of stocks in the portfolio.

Table D.4: Factor portfolios for different sub-periods in Europe

|                                      |                            | R-TC | SD   | SR-TC | Turn. | N  | T    |
|--------------------------------------|----------------------------|------|------|-------|-------|----|------|
| Panel A: Small capitalization stocks |                            |      |      |       |       |    |      |
| 1982-1989                            | Small Market               | 31.6 | 17.1 | 1.19  | 109.1 | 23 | 662  |
|                                      | Value                      | 42.5 | 17.9 | 1.59  | 131.5 | 23 | 118  |
|                                      | Investment                 | 37.8 | 18.3 | 1.36  | 173.3 | 15 | 76   |
|                                      | Profitability              | 31.5 | 18.8 | 1.07  | 153.4 | 18 | 85   |
|                                      | Momentum                   | 46.0 | 21.6 | 1.44  | 309.4 | 8  | 235  |
|                                      | Avg. V/I/P/M               | 46.7 | 20.1 | 1.56  | 213.3 | 11 | 76   |
| 1990-1999                            | Avg. V/I/P   Mom $\geq 50$ | 41.2 | 19.7 | 1.40  | 150.8 | 19 | 68   |
|                                      | Small Market               | 8.7  | 13.1 | 0.26  | 71.6  | 58 | 3150 |
|                                      | Value                      | 10.5 | 14.6 | 0.35  | 85.7  | 38 | 636  |
|                                      | Investment                 | 8.2  | 14.2 | 0.21  | 167.1 | 19 | 526  |
|                                      | Profitability              | 9.9  | 14.5 | 0.31  | 109.3 | 28 | 398  |
|                                      | Momentum                   | 25.9 | 17.2 | 1.07  | 299.5 | 8  | 497  |
| 2000-2009                            | Avg. V/I/P/M               | 17.0 | 15.1 | 0.72  | 222.0 | 11 | 472  |
|                                      | Avg. V/I/P   Mom $\geq 50$ | 14.2 | 16.0 | 0.53  | 180.2 | 10 | 61   |
|                                      | Small Market               | 7.6  | 20.1 | 0.23  | 58.3  | 60 | 5094 |
|                                      | Value                      | 14.2 | 19.1 | 0.55  | 80.6  | 36 | 1143 |
|                                      | Investment                 | 7.5  | 19.4 | 0.23  | 152.1 | 21 | 1075 |
|                                      | Profitability              | 9.1  | 20.9 | 0.28  | 100.7 | 26 | 582  |
| 2010-2018                            | Momentum                   | 7.4  | 21.5 | 0.20  | 327.8 | 7  | 706  |
|                                      | Avg. V/I/P/M               | 18.5 | 17.6 | 0.81  | 227.2 | 10 | 672  |
|                                      | Avg. V/I/P   Mom $\geq 50$ | 17.5 | 17.8 | 0.75  | 126.0 | 23 | 648  |
|                                      | Small Market               | 9.3  | 16.3 | 0.53  | 49.9  | 63 | 4591 |
|                                      | Value                      | 8.7  | 18.1 | 0.44  | 84.7  | 40 | 1030 |
|                                      | Investment                 | 4.8  | 17.9 | 0.24  | 155.6 | 22 | 1092 |
| Panel B: Large capitalization stocks |                            |      |      |       |       |    |      |
| 1982 - 1989                          | Large Market               | 25.5 | 17.2 | 0.90  | 20.6  | 39 | 698  |
|                                      | Value                      | 32.3 | 17.5 | 1.19  | 52.7  | 28 | 101  |
|                                      | Investment                 | 31.7 | 20.1 | 1.02  | 132.0 | 14 | 75   |
|                                      | Profitability              | 29.6 | 18.7 | 1.00  | 57.7  | 23 | 69   |
|                                      | Momentum                   | 29.8 | 20.1 | 0.94  | 339.6 | 7  | 113  |
|                                      | Avg. V/I/P/M               | 32.6 | 19.3 | 1.10  | 198.8 | 11 | 72   |
| 1990-1999                            | Avg. V/I/P   Mom $\geq 50$ | 33.3 | 18.2 | 1.19  | 107.8 | 21 | 56   |
|                                      | Large Market               | 13.5 | 13.3 | 0.58  | 16.9  | 51 | 1178 |
|                                      | Value                      | 15.6 | 15.8 | 0.61  | 54.1  | 33 | 215  |
|                                      | Investment                 | 12.8 | 13.7 | 0.52  | 135.1 | 14 | 165  |
|                                      | Profitability              | 15.7 | 14.4 | 0.68  | 50.6  | 29 | 129  |
|                                      | Momentum                   | 20.3 | 18.1 | 0.75  | 339.4 | 7  | 167  |
| 2000-2009                            | Avg. V/I/P/M               | 19.7 | 15.3 | 0.86  | 218.7 | 10 | 141  |
|                                      | Avg. V/I/P   Mom $\geq 50$ | 17.6 | 15.1 | 0.76  | 110.9 | 21 | 120  |
|                                      | Large Market               | 5.0  | 15.8 | 0.13  | 13.0  | 56 | 1412 |
|                                      | Value                      | 9.6  | 20.1 | 0.32  | 60.1  | 30 | 251  |
|                                      | Investment                 | 11.3 | 17.7 | 0.45  | 125.9 | 14 | 142  |
|                                      | Profitability              | 4.3  | 17.0 | 0.08  | 52.8  | 27 | 95   |
| 2010-2018                            | Momentum                   | 5.5  | 20.7 | 0.12  | 358.9 | 7  | 212  |
|                                      | Avg. V/I/P/M               | 11.4 | 17.1 | 0.47  | 254.6 | 9  | 153  |
|                                      | Avg. V/I/P   Mom $\geq 50$ | 8.5  | 18.7 | 0.29  | 124.1 | 19 | 129  |
|                                      | Large Market               | 5.1  | 14.0 | 0.34  | 9.6   | 62 | 1655 |
|                                      | Value                      | 5.4  | 23.8 | 0.21  | 43.8  | 37 | 189  |
|                                      | Investment                 | 7.2  | 18.3 | 0.36  | 129.8 | 15 | 167  |
|                                      |                            |      |      |       |       |    |      |
|                                      | Profitability              | 8.7  | 14.5 | 0.56  | 33.4  | 38 | 97   |
|                                      | Momentum                   | 11.5 | 18.2 | 0.59  | 359.1 | 7  | 218  |
|                                      | Avg. V/I/P/M               | 9.4  | 18.4 | 0.47  | 277.5 | 9  | 168  |
|                                      | Avg. V/I/P   Mom $\geq 50$ | 10.1 | 19.3 | 0.48  | 114.6 | 20 | 135  |
|                                      |                            |      |      |       |       |    |      |
|                                      |                            |      |      |       |       |    |      |

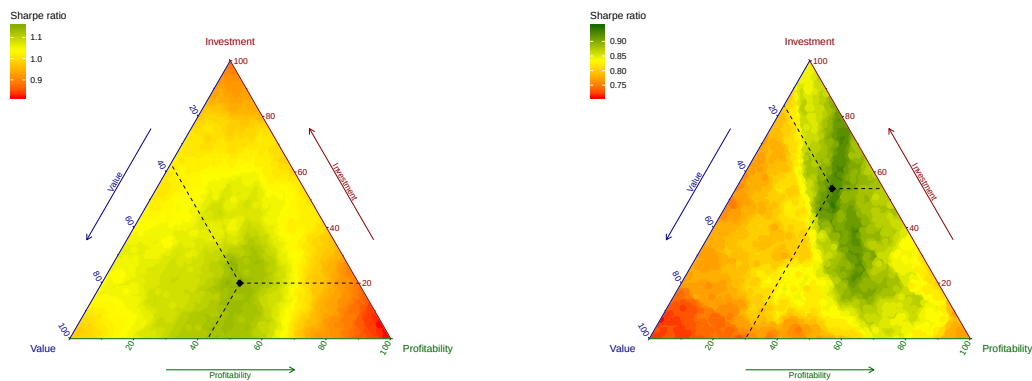
This table shows performance statistics for factor portfolios during different sub periods in Europe. Columns depict the annualized after transaction cost return, standard deviation, Sharpe ratio, turnover, average monthly holding period, and average number of stocks in the portfolio.

Table D.5: Factor portfolios with limited portfolio size in Europe

|                                      | N = unlimited |      |       |       |     | N = 100 |      |       |       |    |
|--------------------------------------|---------------|------|-------|-------|-----|---------|------|-------|-------|----|
|                                      | R-TC          | SD   | SR-TC | Turn. | T   | R-TC    | SD   | SR-TC | Turn. | T  |
| Panel A: Small capitalization stocks |               |      |       |       |     |         |      |       |       |    |
| Small market                         | 13.6          | 16.5 | 0.55  | 106.6 | 105 | 19.5    | 17.2 | 0.82  | 87.9  | 89 |
| Value                                | 19.6          | 16.4 | 0.86  | 127.7 | 50  | 22.7    | 17.2 | 0.98  | 108.4 | 56 |
| Investment                           | 19.9          | 18.5 | 0.78  | 407.1 | 7   | 21.1    | 23.8 | 0.65  | 374.8 | 9  |
| Profitability                        | 14.5          | 17.6 | 0.55  | 191.7 | 23  | 14.5    | 21.0 | 0.46  | 205.0 | 21 |
| Momentum                             | 14.3          | 17.6 | 0.55  | 155.5 | 32  | 14.0    | 19.6 | 0.48  | 161.7 | 29 |
| Avg. V/M/I/P                         | 22.8          | 16.5 | 1.02  | 277.6 | 11  | 27.8    | 17.4 | 1.20  | 213.9 | 17 |
| Avg. V/I/P   Mom $\geq$ 50           | 19.5          | 15.1 | 0.94  | 133.8 | 44  | 19.1    | 14.9 | 0.93  | 100.4 | 70 |
| Panel B: Large capitalization stocks |               |      |       |       |     |         |      |       |       |    |
| Large market                         | 10.9          | 14.6 | 0.45  | 89.6  | 91  | 10.7    | 14.2 | 0.45  | 106.3 | 29 |
| Value                                | 15.4          | 18.6 | 0.57  | 120.4 | 42  | 15.7    | 18.8 | 0.58  | 98.9  | 56 |
| Investment                           | 17.3          | 19.1 | 0.64  | 402.9 | 7   | 17.7    | 20.2 | 0.62  | 339.7 | 8  |
| Profitability                        | 14.3          | 17.9 | 0.54  | 200.8 | 15  | 14.3    | 18.1 | 0.54  | 187.6 | 16 |
| Momentum                             | 14.4          | 17.1 | 0.57  | 119.5 | 38  | 14.5    | 17.1 | 0.58  | 111.6 | 40 |
| Avg. V/M/I/P                         | 18.9          | 17.9 | 0.76  | 298.0 | 10  | 19.2    | 18.1 | 0.77  | 257.2 | 11 |
| Avg. V/I/P   Mom $\geq$ 50           | 16.0          | 16.2 | 0.69  | 124.0 | 48  | 15.5    | 15.9 | 0.68  | 105.9 | 55 |

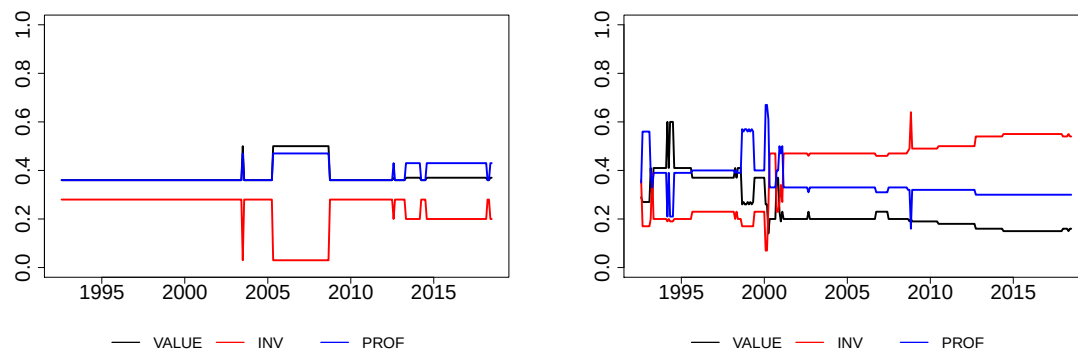
This table shows performance statistics for factor portfolios with a limited portfolio size in Europe. Columns depict the annualized after transaction cost return, standard deviation, Sharpe ratio, turnover, average monthly holding period, and average number of stocks in the portfolio.

Figure D.2: Optimal in-sample weights for conditional multivariate portfolios in Europe  
 (a) Small capitalization stocks  
 (b) Large capitalization stocks



This figure shows heatmaps for the optimal combinations of factor scores for the conditional scoring approach in Europe. The black dot shows the optimal score combination.

Figure D.3: Optimal out-of-sample weights for conditional multivariate portfolios in Europe  
 (a) Small capitalization stocks (b) Large capitalization stocks



These figures display optimal out-of-sample characteristic weights for multivariate factor portfolios. The sample period is July 1982 to June 2018.